

Supplementary Materials for paper: CPDT: A Novel Cluster-based Paired Decision Tree for Identifying Biomedical Entity Interactions

S1. Ensemble Pattern for CPDT

To further enhance CPDT's generalization ability, we introduce an ensemble pattern based on the random forest framework. This approach combines multiple CPDT models, each trained on different feature pair subsets and sample subsets, resulting in a forest model that demonstrates superior performance.

That is, feature pairs from the dataset are sampled without replacement, generating a diverse set of feature pair subsets. Given the scarcity of valuable samples and the common imbalance in class distribution, we address this issue by calculating sampling weight w_i based on the proportion r_i of the i -th class samples in the dataset. This enables the model to perform biased sampling with replacement, thereby achieving improved recognition accuracy for minority class samples. The sampling weight is given in (1).

$$w_i = \frac{1/r_i}{\sum_{i=1}^{N_C} 1/r_i} \quad (1)$$

This process generates a sample subset with balanced classes, which is then used to train a CPDT. This procedure repeats L times to create an ensemble model. Weighted prediction is then applied to obtain its final output. The proposed ensemble pattern effectively leverages the diversity of feature pair subsets and sample subsets, thereby enhancing the performance of the original CPDT.

S2. Information about the Employed Evaluation Metrics

As mentioned in the main text, we employed four evaluation metrics (i.e. ACC , $F1$, AUC and $AUPR$) for performance comparison, as shown in (2-7). Here TP/FP indicates the positive predictions are right/wrong, and TN/FN indicates the negative predictions are right/wrong. REC indicates the accuracy of the classifier's prediction of positive samples, and PRE indicates the proportion of positive samples in the positive prediction. $F1$ is a measure that considers both PRE and REC , maintaining a reasonable measurement on imbalanced datasets. AUC is a performance index to measure the quality of classifiers, where $\sum I(P_{pos}, P_{neg})$ represents the number of positive samples with higher prediction probability than negative samples, and N_{pos}/N_{neg} represents the number of positive/negative samples. It is not affected by the classification threshold and reflects the overall prediction ability of a model. $AUPR$ measures the performance of a binary classification model at different thresholds, which calculates the average precision at different recall levels, and we use the trapezoidal rule to estimate the area under the curve.

$$ACC = \frac{TP+TN}{TP+TN+FP+FN} \quad (2)$$

$$REC = \frac{TP}{TP+FN} \quad (3)$$

$$PRE = \frac{TP}{TP+FP} \quad (4)$$

$$F1 = \frac{2*PRE*REC}{PRE+REC} \quad (5)$$

$$AUC = \frac{\sum I(P_{pos}, P_{neg})}{N_{pos} * N_{neg}} \quad (6)$$

$$AUPR = \sum_{k=1}^{N_S} (REC_k - REC_{k-1}) * PRE_k \quad (7)$$

S3. Analysis of Decision Boundaries for Partitioning Methods

Fig. S1 shows the data distribution in a specific paired feature space, along with the sample partitioning scheme of various sample partitioning methods. The horizontal and vertical axes represent the expression values of entities A and B in the paired feature space, respectively. Fig. S1(A) shows the real label distribution of the data, with dark blue and dark red points denoting samples from different classes.

Fig. S1(B) and Fig. S1(C) depict the outcomes of sample partitioning using K-means and DBSCAN, respectively. Since K-means can only identify convex-shaped clusters, it fails to assign samples in class 1 to the same cluster and wrongly mixes the two classes. As a density-based clustering method, DBSCAN can recognize closely connected samples in non-convex shapes, but it splits more clusters than expected. Furthermore, both clustering methods don't provide their decision boundaries, thus lacking interpretability in quantifying the role of features in decision-making.

Fig. S1(D) and Fig. S1(E) illustrate the outcomes of conventional methods, where entity features are treated independently, on each dimension of the paired feature space. Concerning the process of sample partitioning in decision trees, we determine the split point that maximizes information gain to establish the decision boundary. As depicted in the figure, while introducing a simple axis-parallel decision boundary enhances interpretability, it may not accommodate complex data distributions. Although repeated partitioning may yield reasonably suitable clusters, each partitioning step often results in some loss of information about the data distribution. Specifically, neighboring data points in the distribution may be assigned into separate subgroups, discarding their spatial connection. In interaction prediction, the focus should be on the molecules corresponding to feature pairs that influence the output of an entity pair, rather than on the individual features within a single entity. For example, in the field of drug combination prediction, researchers are primarily interested in understanding how the transcriptional expression levels of specific genes are altered following the administration of drug combinations, rather than solely examining the isolated regulatory effects of individual drugs on those genes.

Fig. S1(F) depicts the sample partitioning scheme using linear discriminant analysis. By identifying a projection plane that minimizes the distance between homogenous samples and maximizes the separation between heterogeneous samples, this approach achieves significant discrimination performance. Its decision boundary can be expressed as a binary first-order inequality, e.g. $w_A x^{A,j} + w_B x^{B,j} > t$, where w_A and w_B denote the weights of two features in the paired feature space, and t represents the decision threshold. However, this form is not axis-parallel and lacks biological interpretability, making the task complicated for experts attempting to interpret the decision rules.

Fig. S1(G) shows the outcomes of sample partitioning using the proposed adaptive grid-based clustering algorithm. Compared with Fig. S1(D-F), our method minimizes the information loss of data distribution during the node split process by employing a multi-segment partitioning scheme. This approach flexibly splits the paired feature space into adaptive-scale grids that are subsequently merged. Additionally, it preserves the interpretability of its decision rules in the form of multiple simple axis-parallel decision boundary combinations, making the partitioning results more understandable.

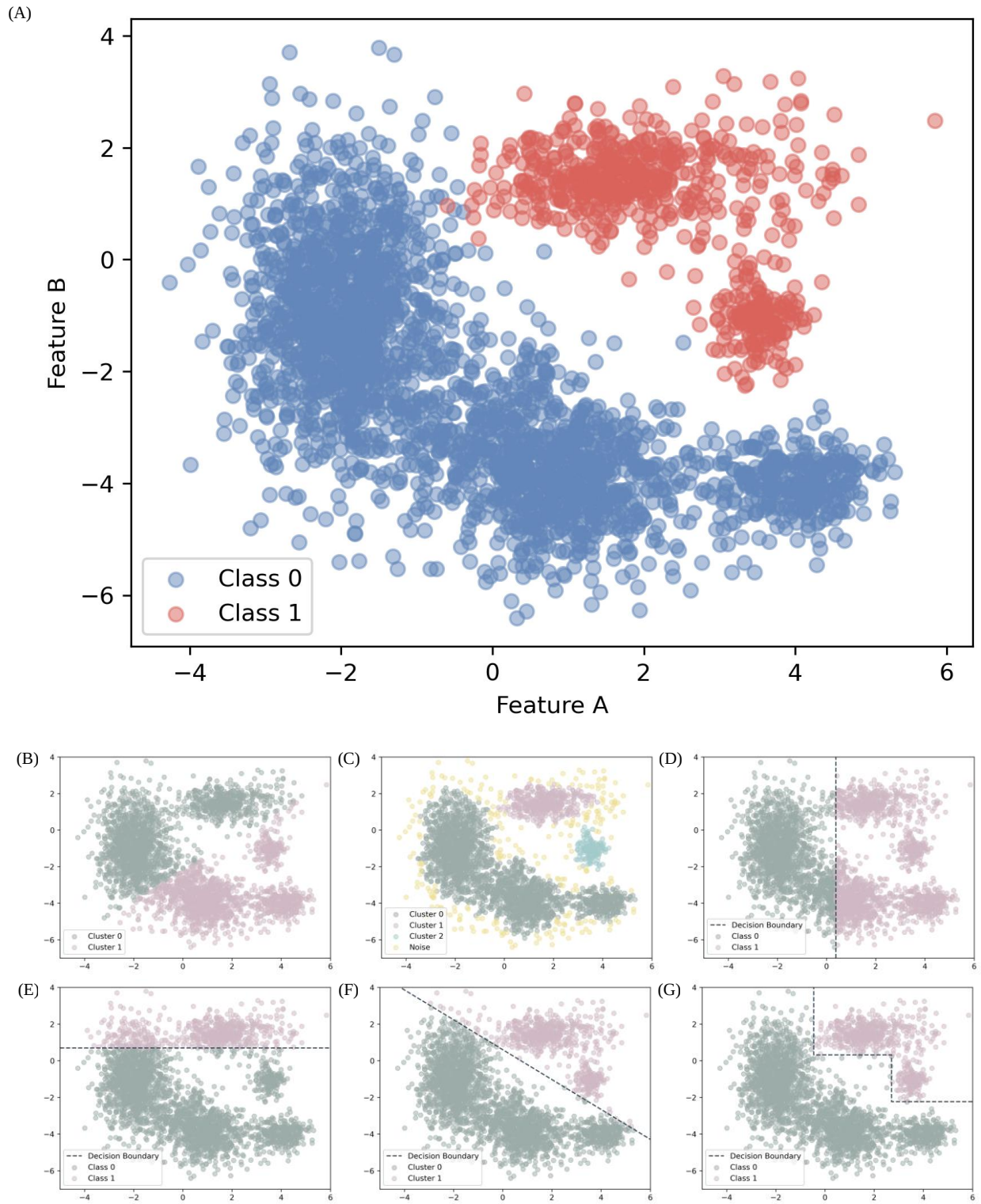


Fig. S1. The real data distribution in the paired feature space (A) and the sample partitioning scheme of various methods (B-G).

S4. Results of feature importance Analysis for CPDT

In the left fan plot of Fig. S2, we have categorized the feature pairs into four statistical intervals: [1, 10], [11, 50], [51, 300], and [301, 978]. This division allows for an intuitive evaluation of the overall contribution of different importance levels to the model. In the scatter plot on the right of Fig. S2, each point represents a feature pair, both the y-axis height and color represent its *FI*, and the red dashed line represents the mean.

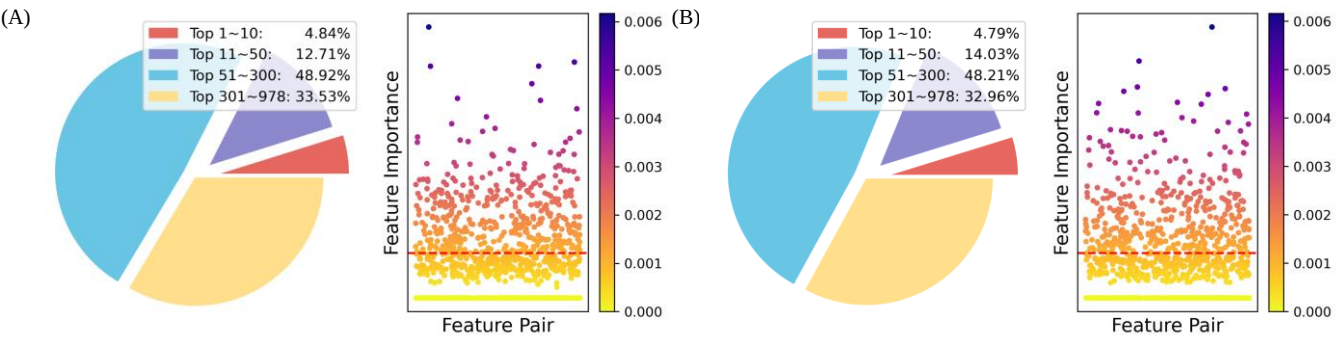


Fig. S2. Feature importance statistics on (A) DC and (B) SL.

TABLE SI. INFORMATION ABOUT THE TOP 10 KEY GENES

DC			SL		
Rank	Name	FI	Rank	Name	FI
1	C2CD2L	0.006169	1	JMJD6	0.006161
2	PLEKHM1	0.005370	2	HSPA8	0.005387
3	MTA1	0.005279	3	HACD3	0.004797
4	GPB1	0.005276	4	NCOA3	0.004760
5	GRWD1	0.004878	5	STK10	0.004699
6	ECD	0.004545	6	MTA1	0.004629
7	PDLIM1	0.004499	7	HEBP1	0.004448
8	CTSL	0.004308	8	C2CD2L	0.004423
9	PPAR	0.004125	9	GRWD1	0.004406
10	SPAG4	0.003967	10	CXCL2	0.004211

S5. Enrichment Results of Key Genes Identified by CPDT

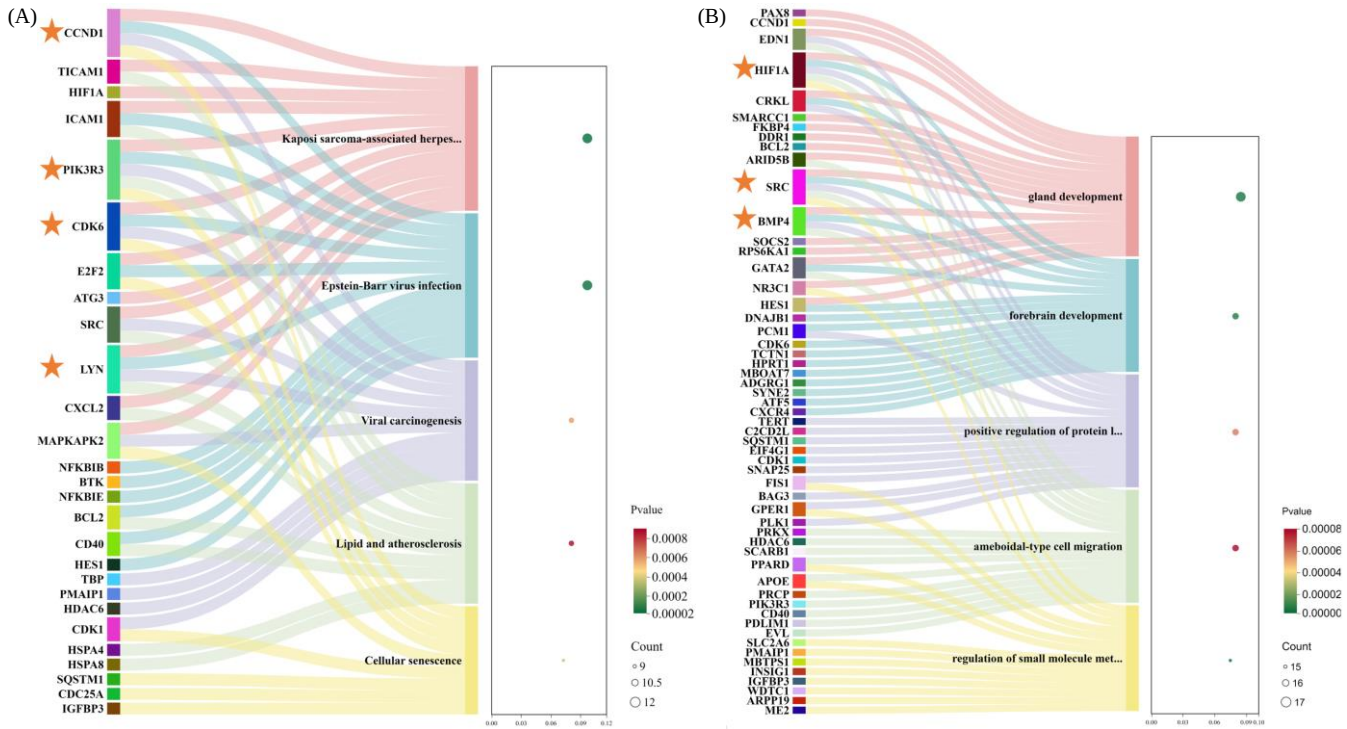


Fig. S3. Enrichment results of key genes in (A) KEGG pathways, (B) gene ontology biological processes.

S6. Information and Prediction about the Novel Drug Combinations

Within the DrugComb database, each drug is paired with others, and samples with labeled synergy scores are excluded. This process results in 432 unmeasured drug combinations across 13 cell lines, forming the prediction set for novel drug combinations. The probability prediction of class 1 generated by CPDT (named *Score*) is ranked to identify drug combinations with high prediction scores, which are considered potential synergistic drug combinations. Details of the top 20 potential synergistic drug combinations are given in Table SII.

TABLE SII. TOP 20 POTENTIAL SYNERGISTIC DRUG COMBINATIONS IDENTIFIED BY CPDT

Rank	Drug A	Drug B	Score	Cell line
1	Exemestane	BEZ-235	0.737871	HT29
2	Etoposide	Nilotinib	0.737699	HT29
3	Etoposide	Tretinoin	0.729155	HT29
4	Lenalidomide	BEZ-235	0.712887	HT29
5	Erlotinib Hydrochloride	BEZ-235	0.701600	HT29
6	Dexrazoxane	BEZ-235	0.701307	HT29
7	Mitotane	BEZ-235	0.700468	HT29
8	Thalidomide	BEZ-235	0.700465	HT29
9	Megestrol Acetate	BEZ-235	0.697121	HT29
10	Etoposide	Lenalidomide	0.695915	HT29
11	Etoposide	Erlotinib Hydrochloride	0.687730	HT29
12	Lenalidomide	MK-2206	0.684887	HT29
13	Etoposide	Dexrazoxane	0.680306	HT29
14	Etoposide	Pazopanib Hydrochloride	0.671088	HT29
15	Exemestane	MK-5108	0.670220	HT29
16	Pazopanib Hydrochloride	BEZ-235	0.670018	HT29
17	Methoxsalen	BEZ-235	0.668740	HT29
18	Etoposide	Ruxolitinib	0.668236	HT29
19	Etoposide	Imiquimod	0.666798	HT29
20	Celecoxib	BEZ-235	0.665977	HT29

In Fig. S4, each circle represents a drug, and the circles are connected by chords, and each chord represents a drug combination. The width of the chord indicates the predicted synergy score of the drug combination. The more chords a circle connects, the more frequently the drug appears in the identified drug combinations.

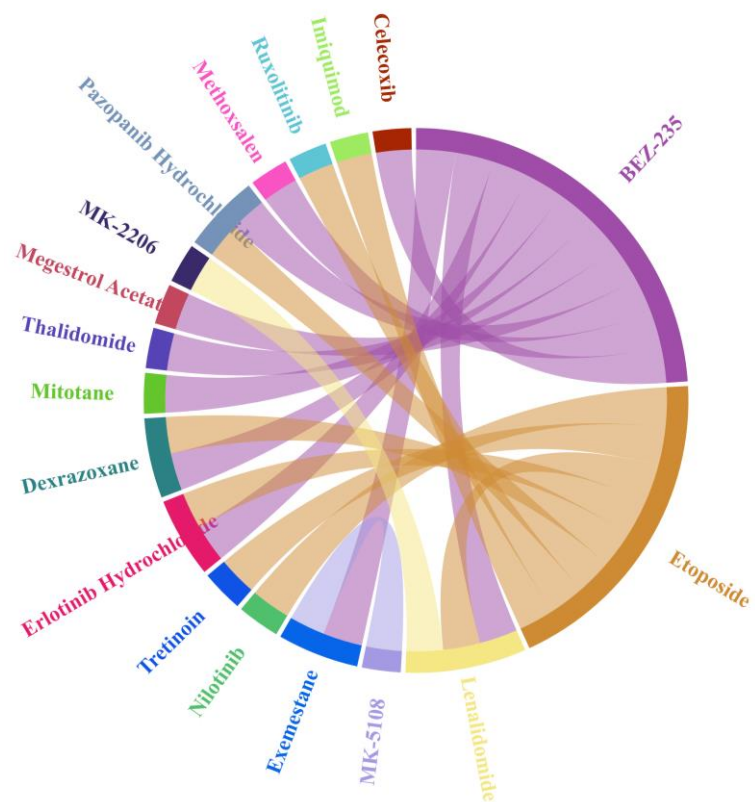


Fig. S4. Relationships between drugs contained in the top 20 potential synergistic drug combinations